UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 4A – Physical Ergonomics: Work Physiology I

EIN 3241: Human Factors and Ergonomics
Spring 2026

February 6, 2025

Dr. Jad A. Atweh

Admin Issues

- Lab 1: February 11
- Quiz 3: February 18

MOI Example

- Calculate the center of mass of the leg and the moment of inertia about its proximal end. The body weight is 80 Kg. The actual leg (shank) length is 0.435 m.

Table 3.2 Dempster's Body Segment Parameters

Segment	Endpoints[a] (proximal to distal)	SEGMENTAL MASS/TOTAL MASS	CENTER OF MASS/SEGMENT LENGTH		RADIUS OF GYRATION/ SEGMENT LENGTH		
		$(P)^b$	$(R_{proximal})^c$	$(R_{distal})^c$	$(K_{cg})^d$	$(K_{proximal})^d$	$(K_{distal})^d$
Hand	Wrist center to knuckle II of third finger	0.0060	0.506	0.494	0.298	0.587	0.577
Forearm	Elbow to wrist center	0.0160	0.430	0.570	0.303	0.526	0.647
Upper arm	Glenohumeral joint to elbow center	0.0280	0.436	0.564	0.322	0.542	0.645
Forearm and hand	Elbow to wrist center	0.0220	0.682	0.318	0.468	0.827	0.565
Upper extremity	Glenohumeral joint to wrist center	0.0500	0.530	0.470	0.368	0.645	0.596
Foot	Ankle to ball of foot	0.0145	0.500	0.500	0.475	0.690	0.690
Leg	Knee to ankle center	0.0465	0.433	0.567	0.302	0.528	0.643
Thigh	Hip to knee center	0.1000	0.433	0.567	0.323	0.540	0.653
Lower extremity	Hip to ankle center	0.1610	0.447	0.553	0.326	0.560	0.650
Head	C7-T1 to ear canal	0.0810	1.000	0.000	0.495	1.116	0.495
Shoulder	Sternoclavicular joint to glenohumeral joint center	0.0158	0.712	0.288			
Thorax	C7-T1 to T12-L1	0.2160	0.820	0.180			
Abdomen	T12-L1 to L4-L5	0.1390	0.440	0.560			
Pelvis	L4-L5 to trochanter	0.1420	0.105	0.895			
Thorax and abdomen	C7-T1 to L4-L5	0.3550	0.630	0.370			
Abdomen and pelvis	T12-L1 to greater trochanter	0.2810	0.270	0.730			
Trunk	Greater trochanter to glenohumeral joint	0.4970	0.495	0.505	0.406	0.640	0.648
Head, arms, and trunk	Greater trochanter to glenohumeral joint	0.6780	0.626	0.374	0.496	0.798	0.621
Head, arms, and trunk	Greater trochanter to mid-rib	0.6780	1.142	−0.142	0.903	1.456	0.914

[a]Endpoints are defined in table 3.1.

[b]A segment's mass as a proportion of the total body mass.

[c]The distances from the proximal and distal ends of the segment to the segment's center of gravity as proportions of the segment's length.

[d]The radii of gyration about the center of gravity, proximal and distal ends of the segment to the segment's center of gravity as proportions of the segment's length.

Adapted from D.A. Winter 1990.

Analysis of Human Activity

- Stress: outside effector
- Strain: change in the body due to stress
- A human activity continuum for categorizing and quantifying strain:

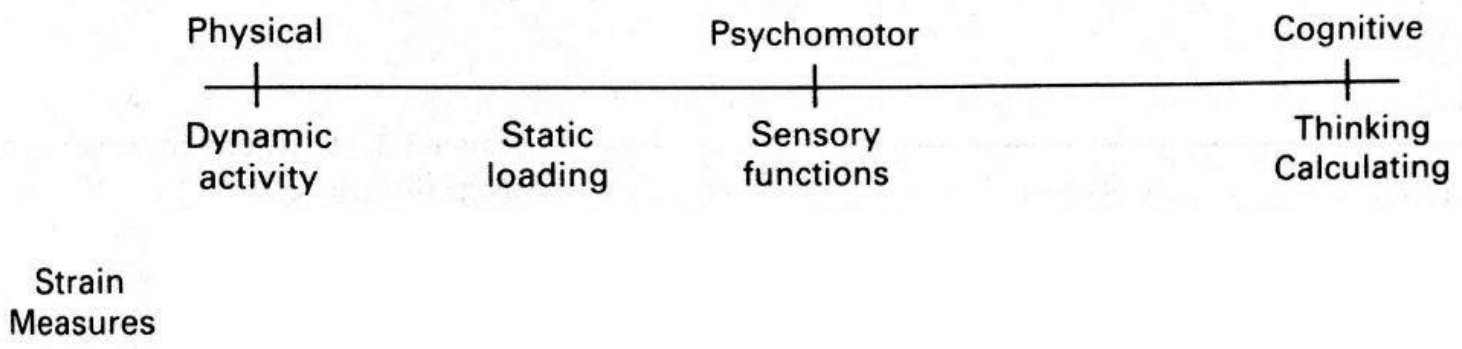
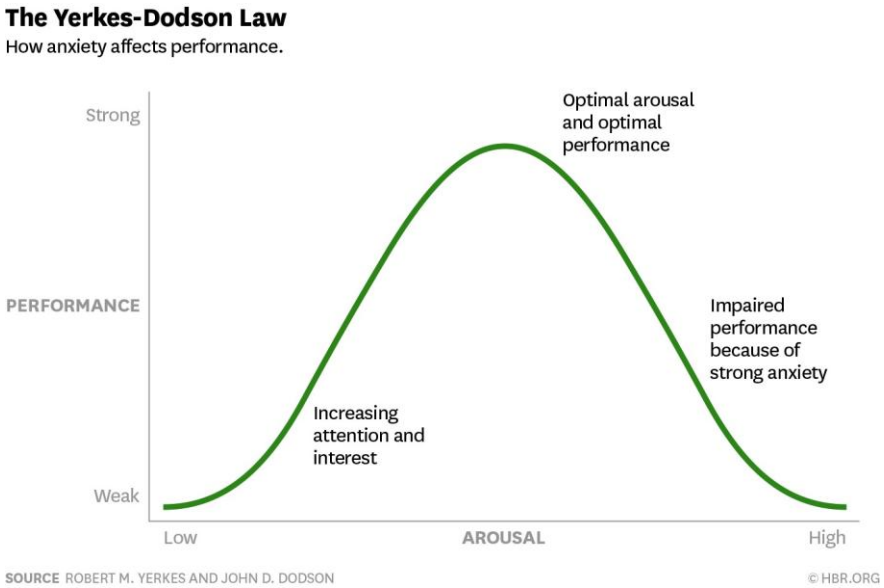
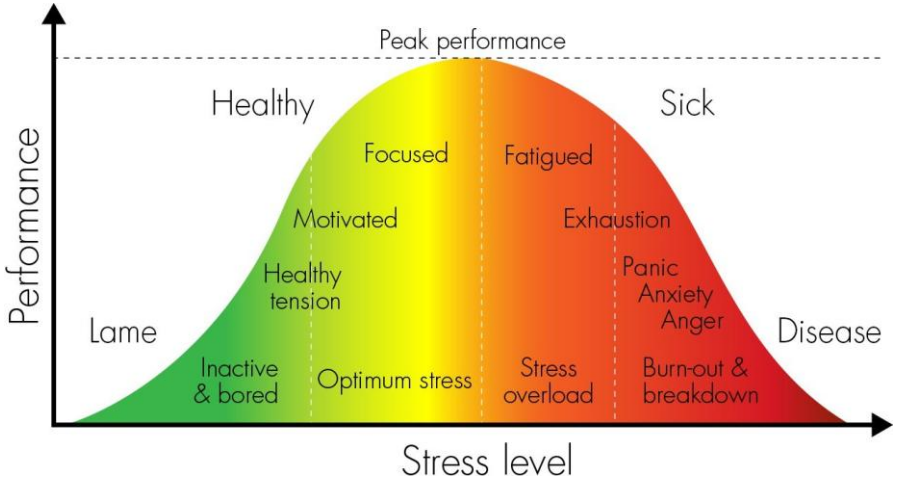


Figure 3.1 Human activity continuum and the primary measures of strain.

Stress and Performance

- Stress and performance curve.



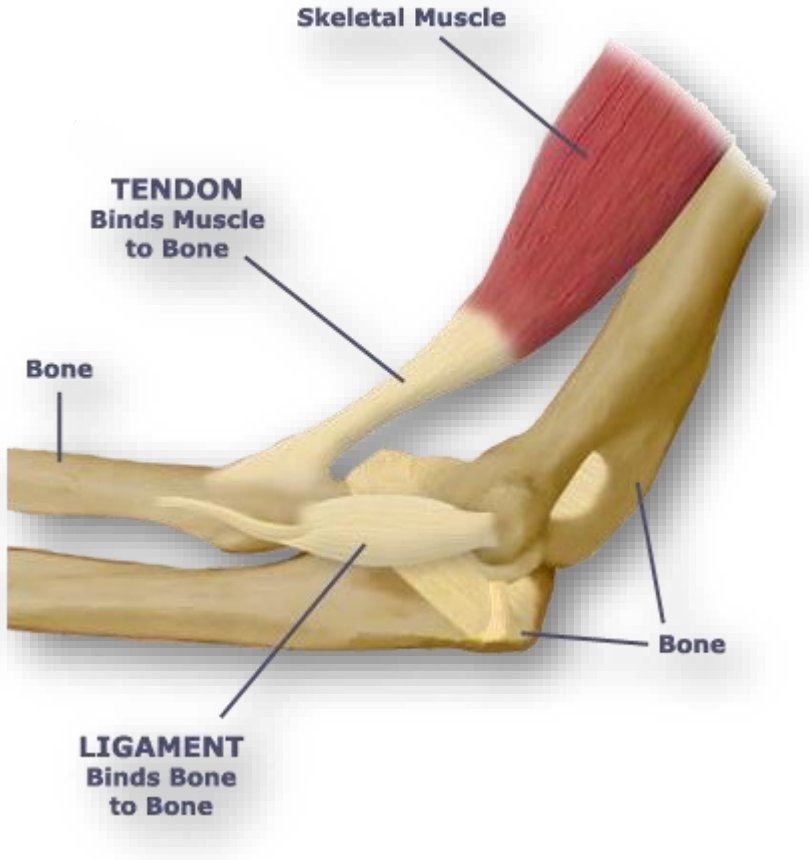


Musculoskeletal System: Bones

- Musculoskeletal system composed of muscles, bones, and connective tissue, which collectively provide muscular activity.
- Bones
  - 206 bones
  - Establish the framework of human structure
  - With muscles, they provide mobility for
  - human activity
  - Cover vital organs (e.g. skull, rib cage)

Musculoskeletal System: Ligaments/Tendons

- Ligaments: connects bone to bone.
- Tendons: connects muscle to bone.



Musculoskeletal System: Muscles

- Three types of muscles:

Smooth muscle

- occurs in walls of internal organs.
- is involuntary.

Cardiac muscle

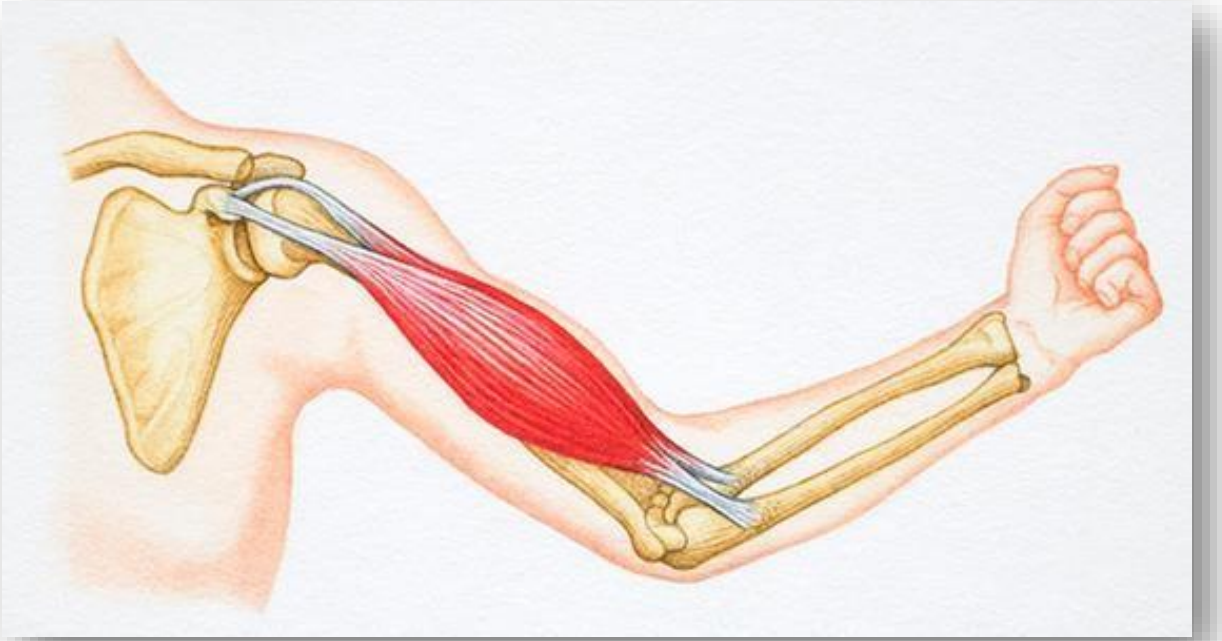
- occurs in walls of heart.
- is involuntary.

Skeletal muscle

- is usually attached to skeleton.
- is voluntary.

Musculoskeletal System: Skeletal Muscles

- Human movement is a result of forces generated by the contraction of skeletal muscles.
- The larger the muscle (its cross section), the larger the forces that can be applied with it.



Musculoskeletal System: Muscles

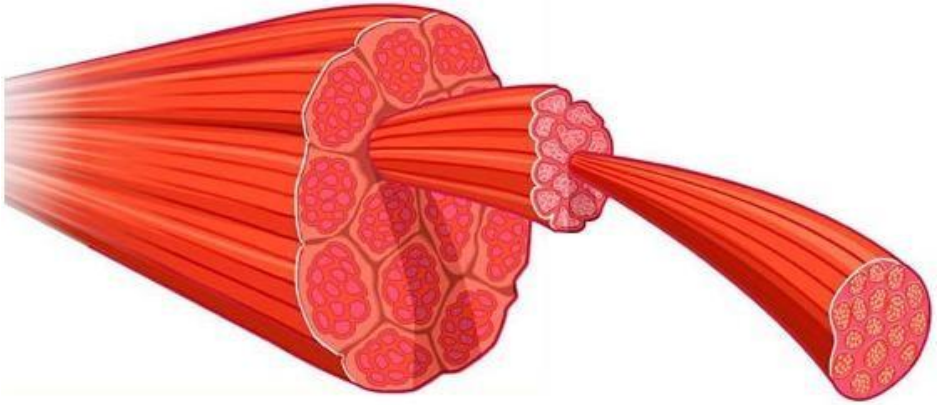
Muscles are composed of bundled **muscle fibers** (i.e. muscle cells).

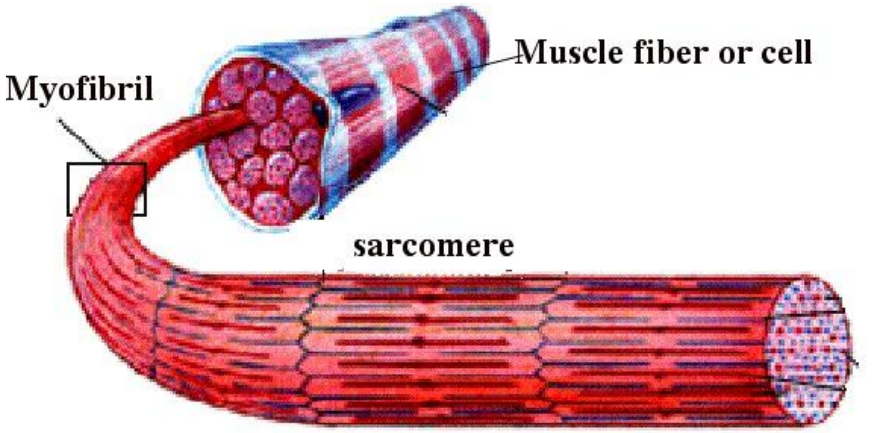
Muscle cells are composed of **myofibrils**.

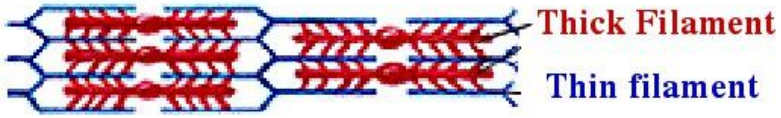
Myofibrils are composed of repeating sections of **sarcomeres**.

Sarcomeres are composed of long chains of protein called **filaments**:

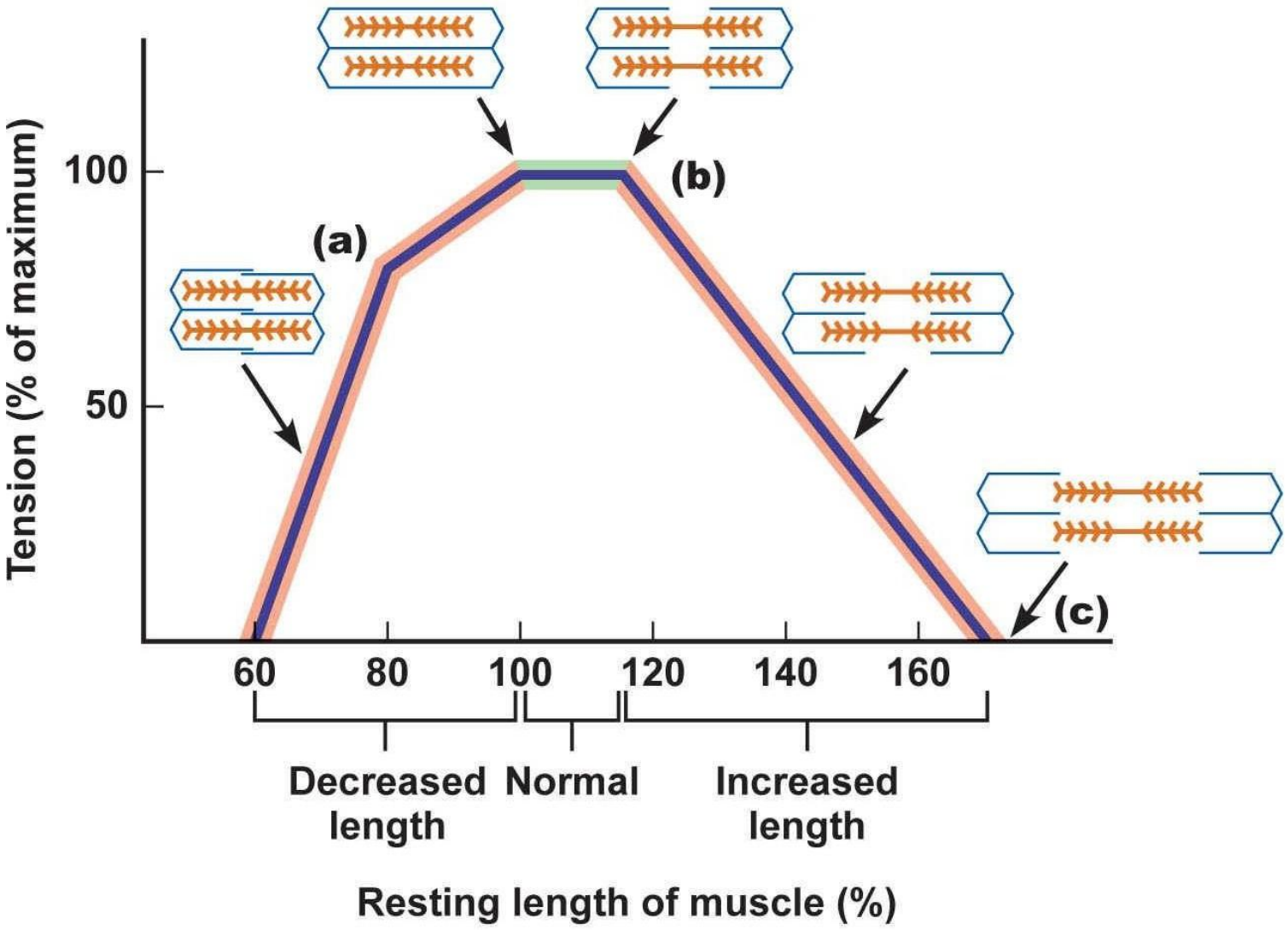
- Thick Filament (Myosin)
- Thin Filament (Actin)







Length-Force Relationship



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Two Types of Muscular Effort

- Ergonomists differentiate between two types of muscular effort:

1. Dynamic Effort:
   - Involves rhythmic contraction and relaxation of muscles.
   - Allows more blood to be circulated through the muscle, effectively meeting oxygen and waste removal needs.
2. Static Effort:
   - Involves a prolonged state of contraction in a fixed position.
   - Restricts blood flow to the muscle tissue.

Energy Cost of Work: EMG

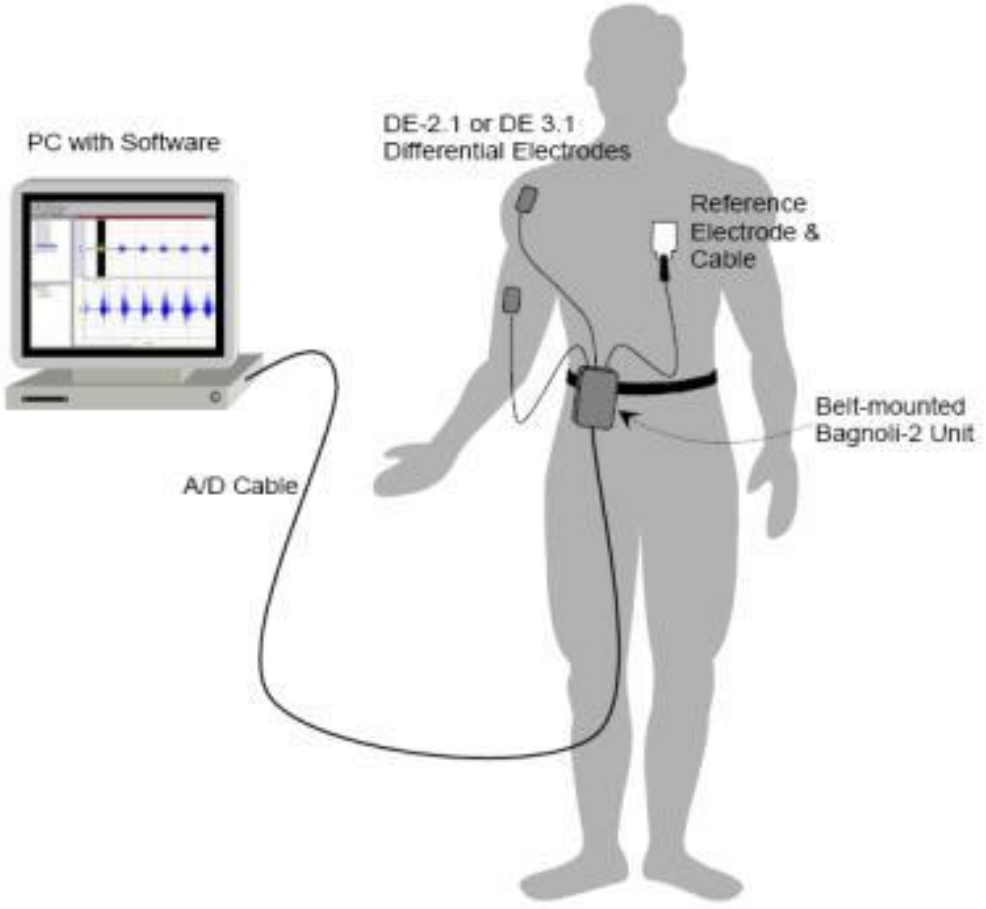
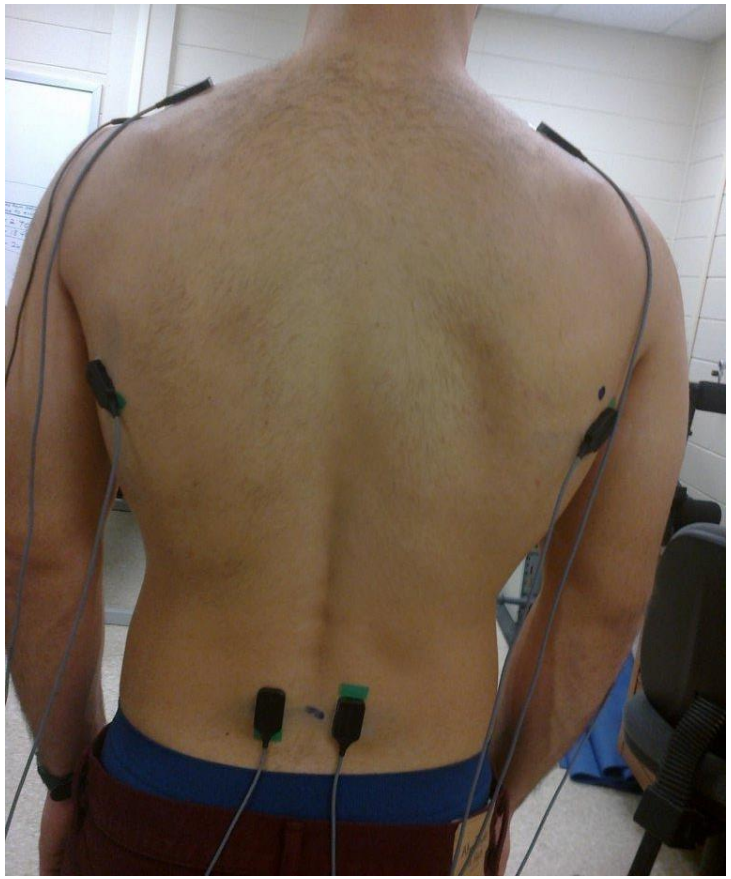


Figure 1: Components of the Bagnoli-2 EMG System.



Energy Cost of Work: EMG

- EMG answers questions such as:

Level of question	Type of answer	Type of scaling
1) Is the muscle active?	Yes/Noand On/Off	Nominal
2) Is the muscle more or less active?	Ranking between tests in qualitative terms	Ordinal
3) When is the muscle on/off?	Onset/Offset calculations, firing orders	Metric
4) How much is the muscle active?	Expressed in e.g. % MVC	Metric
5) Does the muscle fatigue?	Slope calculation of EMG parameters	Metric

- Drawback: No clear-cut definition of muscular work limit with local loading.

Musculoskeletal System: Metabolism

- Metabolism is the chemical process of conversion of food into mechanical work and heat.
- Metabolism never stops. It supplies the necessary energy for breathing, blood circulation, digesting food, cell mechanics, regulating body temperature, and functionality of the musculoskeletal system.

Musculoskeletal System: Metabolism

- Aerobic work: If sufficient oxygen is supplied to the muscle, the pyruvic acid is broken down into water and carbon dioxide, releasing large amounts of ATP.

$$\text{glucose} + 38\ \text{phosphate} + 38\ \text{ADP} + 6\ O_2 \rightarrow 6\ CO_2 + 44\ H_2O + 38\ \text{ATP}$$

- Anaerobic work: Under insufficient oxygen supply, pyruvic acid is converted into lactic acid while ATP is regenerated. Latic acid accumulation between muscle fibers causes muscular fatigue and pain.

$$\text{glucose} + 2\ \text{phosphate} + 2\ \text{ADP} \rightarrow 2\ \text{lactate} + 2\ \text{ATP}$$

Musculoskeletal System: Metabolism

- Kilocalorie (kcal): measure of energy requirement for physical activity.
- The average adult needs 2,000 – 2,500 kcal/day.
- *Total Metabolism = Basal Metabolism + Digestive Metabolism + Activity Metabolism*
- **Basal Metabolism**: the minimum steady energy consumption to maintain involuntary activity (excluding digestion).
  - For males (70 kg) about 1700 kcal/day
  - For females (60 kg) about 1400 kcal/day
- **Digestive metabolism**: The amount of energy consumed by the digestive system. Can be estimated as 10% of the sum of basal and activity metabolisms.
- **Activity Metabolism:** The amount of energy is required to perform daily or work activities.

BM Formula

- Predictive Basal Metabolism equation, first developed by Harris-Benedict (1919). It has been later modified.

- New predictive Basal Energy Expenditure equation, developed by Mifflin (1990):

- BM in SI:

$$BM_{men}(kcal) = 10 \times Weight(kg) + 6.25 \times Height(cm) - 5 \times Age(years) + 5$$

$$BM_{women}(kcal) = 10 \times Weight(kg) + 6.25 \times Height(cm) - 5 \times Age(years) - 161$$

- BM in US:

$$BM_{men}(kcal) = 4.536 \times Weight(lb) + 15.88 \times Height(in) - 5 \times Age(years) + 5$$

$$BM_{women}(kcal) = 4.536 \times Weight(kg) + 15.88 \times Height(in) - 5 \times Age(years) - 161$$

AM Formula

$$AM_{men}(kcal) = \frac{T \times (0.6309 \times HR - 0.1988 \times W + 0.2017 \times A - 55.0969)}{4.184}$$

$$AM_{women}(kcal) = \frac{T \times (0.4472 \times HR - 0.1263 \times W + 0.074 \times A - 20.4022)}{4.184}$$

$Where:$

$T = Duration\ of\ the\ exercise, in\ minutes$

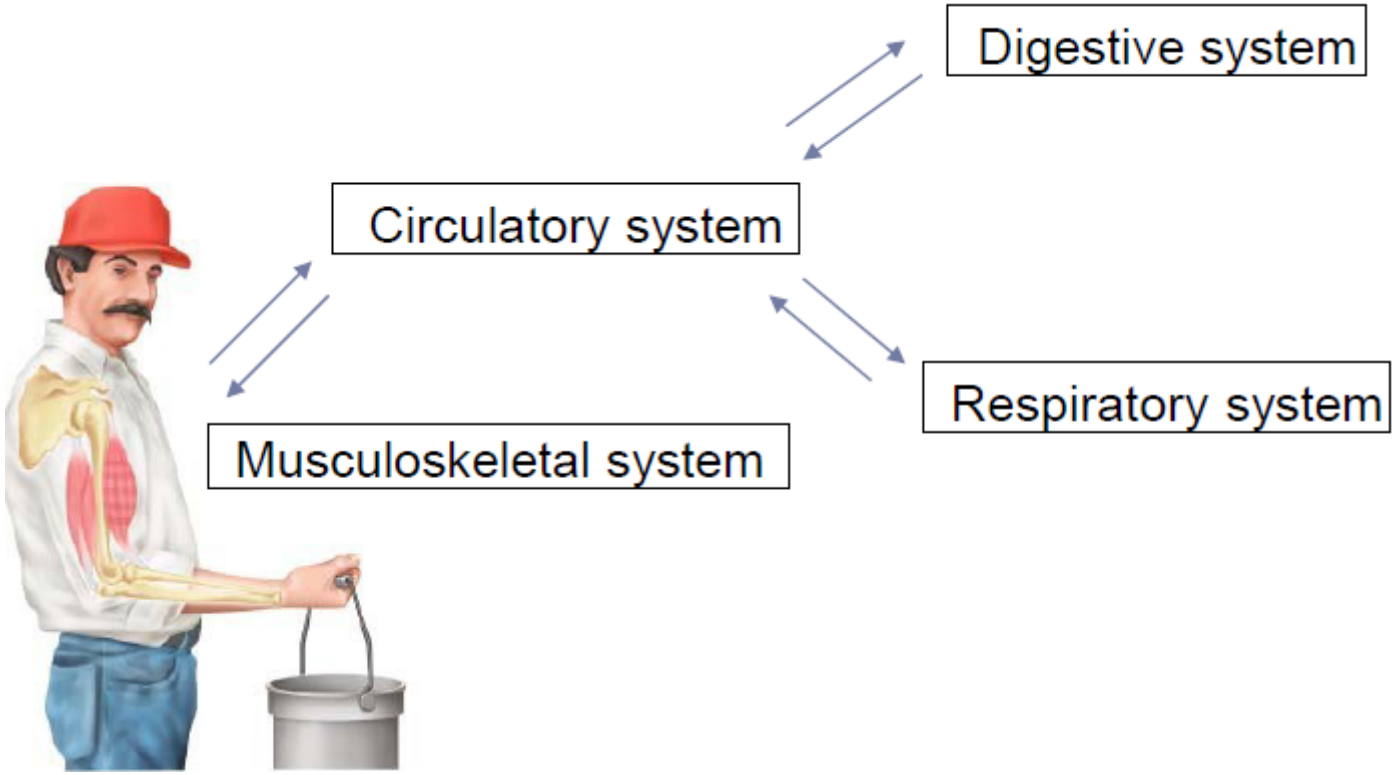
$HR = Average\ hear\ rate, in\ bpm$

$W = Weight\ in\ kilograms$

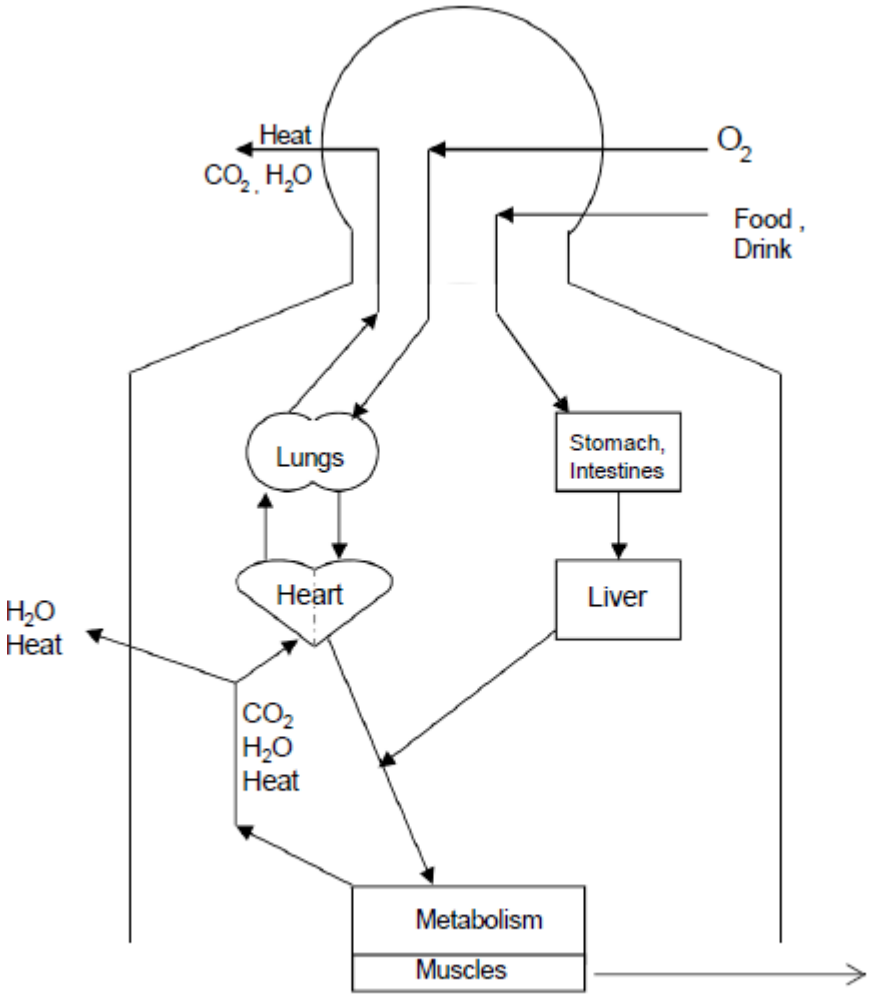
$A = Age\ in\ years$

- Use this equation for heart rate above 115 bpm

Supporting Systems



Supporting Systems



Ref: Kroemer et al, 1990

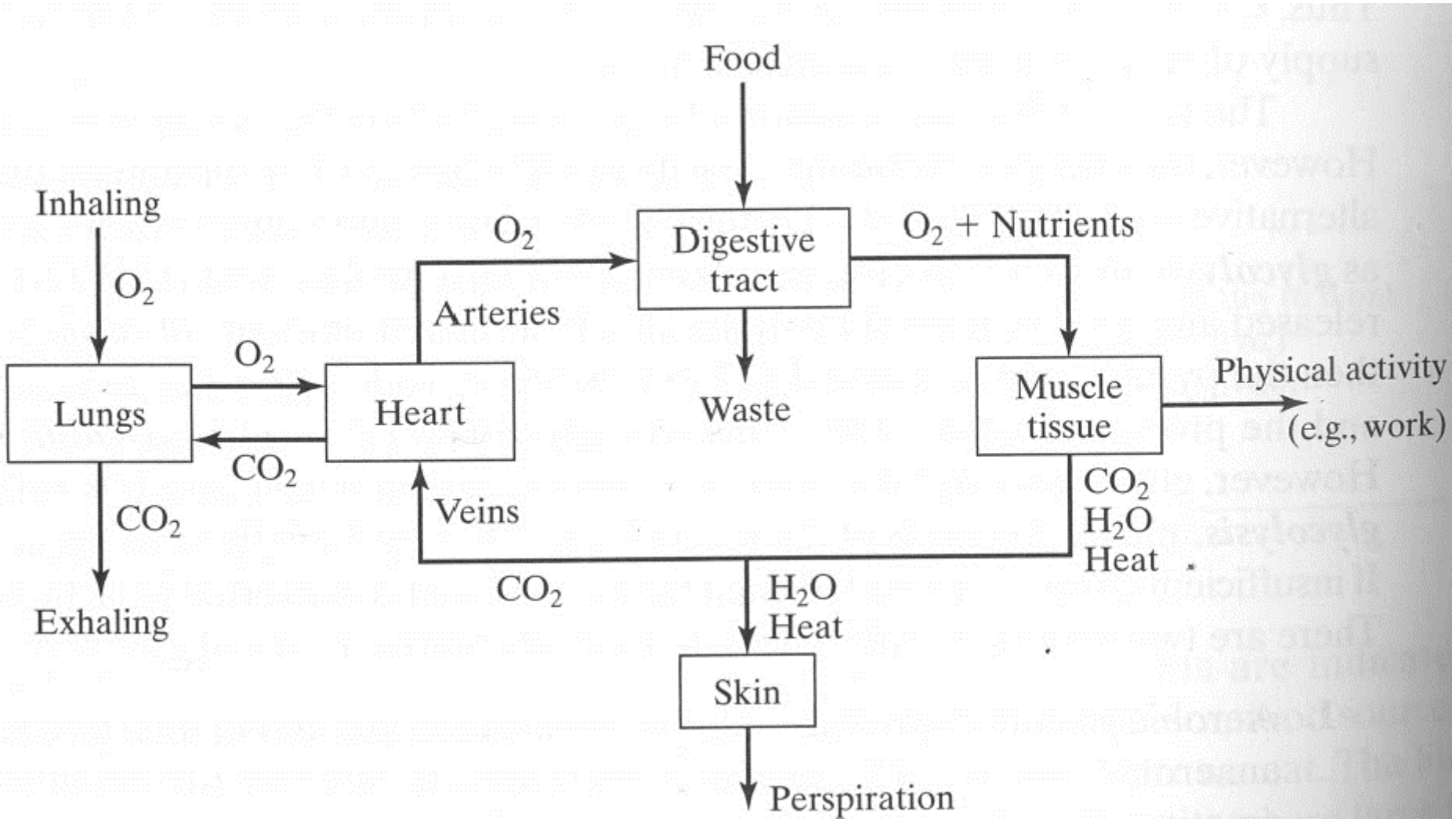
- Breathing accelerate to oxygenate more blood per unit time in the lungs
- Oxygen consumption increases

- More blood is pumped to the affected areas
- Heart rate increases since more blood is processed per unit time

- Energy demands increases → need more oxygen and remove the accumulated waste

- Onset of physical work → muscle contracts

Supporting Systems



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College of Engineering
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LEADING THE CHARGE, CHARGING AHEAD